

AI-Based Smart Traffic Monitoring and Violation Detection System

Short Presentation Script for 3 Members

MEMBER 1 (Slides 1–6)

Slide 1 – Introduction

Good morning everyone. Our project is AI-Based Smart Traffic Monitoring and Violation Detection System. This system uses Artificial Intelligence and CCTV cameras to detect traffic violations automatically.

Slide 2 – Introduction

Traffic accidents are increasing due to overspeeding and wrong-way driving. Manual monitoring is difficult and inefficient, so we developed an AI-based smart monitoring system.

Slide 3 – Problem Statement

Current traffic systems require continuous human monitoring and expensive hardware like radar systems. Our project provides a low-cost smart solution using normal CCTV cameras.

Slide 4 – Objectives

The main objectives of our project are detecting overspeeding vehicles, wrong-way driving, number plate recognition, and sending instant alerts automatically.

Slide 5 – System Architecture

The camera captures live traffic video. YOLO detects vehicles, ByteTrack tracks them, OpenCV analyzes speed and direction, and EasyOCR reads number plates.

Slide 6 – Technologies Used

We used Python, YOLOv8, OpenCV, ByteTrack, EasyOCR, and Telegram API to build the complete intelligent traffic monitoring system.

MEMBER 2 (Slides 7–12)

Slide 7 – YOLOv8 Detection

YOLOv8 detects vehicles such as cars, buses, trucks, and bikes in real time with high accuracy.

Slide 8 – ByteTrack Tracking

ByteTrack assigns unique IDs to vehicles and tracks their movement frame by frame for accurate monitoring.

Slide 9 – Speed Detection

Vehicle speed is calculated using virtual lines and frame timing. This provides a low-cost alternative to radar systems.

Slide 10 – Wrong-Way Detection

The system checks vehicle direction continuously. If a vehicle moves in the wrong direction, the system instantly generates a violation alert.

Slide 11 – ANPR OCR

EasyOCR extracts the vehicle number plate automatically from captured images for identification purposes.

Slide 12 – Automated Alerts

The system stores violation details and sends instant Telegram alerts to traffic officers along with evidence images.

MEMBER 3 (Slides 13–18)

Slide 13 – Workflow

The complete workflow is: Vehicle Detection, Vehicle Tracking, Speed Analysis, OCR Processing, and Alert Generation.

Slide 14 – Advantages

Our system is low-cost, real-time, accurate, scalable, and reduces manpower requirements.

Slide 15 – Applications

This system can be used in smart cities, highways, traffic junctions, and toll plazas.

Slide 16 – Future Scope

Future improvements include helmet detection, seatbelt detection, accident prediction, and cloud integration.

Slide 17 – Conclusion

Our AI-based system improves traffic monitoring, road safety, and supports smart city infrastructure.

Slide 18 – Thank You

Thank you everyone. We are ready to answer your questions.
